

Kevin Pierce Ramos

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Software engineer with hands-on production experience across embedded systems, application development, and cloud infrastructure. Professional background includes C++/Python development for robotic manipulation systems, hardware integration and diagnostics on Linux platforms, and building an operator-facing Android control application in Kotlin. Project experience spans cloud-native REST APIs with CI/CD pipelines, network security appliances, and systems-level tooling. Holds a B.S. in Computer Science from UNLV and begins an M.S. in Computer Science at Georgia Tech in Spring 2027.

PROFESSIONAL EXPERIENCE

Richtech Robotics, Las Vegas, NV

Oct 2025 – Present

Software Engineer

- Engineered and supported dual 6-DOF robotic manipulation systems in live production environments, developing C++ and Python automation logic for motion control, calibration routines, and hardware diagnostics across Linux-based platforms.
- Contributed to a Kotlin Multiplatform and Jetpack Compose Android tablet application that serves as the operator control interface for a Richtech robot, building UI screens, networking, and business logic using Retrofit/OkHttp, a repository pattern, Koin dependency injection, and MVVM ViewModels.
- Integrated and troubleshoot industrial serial communication interfaces (RS-232/RS-485) connecting servo drives, motor controllers, and distributed hardware components; performed system bring-up, signal validation, and fault diagnosis.
- Built observability and monitoring tooling from scratch, including structured logging, real-time metrics collection, and fault alerting, to surface hardware and software performance issues proactively; collaborated with mechanical and electrical engineers on integration and delivery.

PROJECTS

Event Analytics API · *FastAPI, PostgreSQL, Docker, Terraform, GitHub Actions, AWS EC2/S3, SQLAlchemy, Alembic, pytest*

- Designed and built a production-ready REST API featuring time-series analytics, filtered queries, pagination, and schema migration via Alembic. Containerized with Docker, provisioned AWS infrastructure with Terraform, and implemented a full GitHub Actions CI/CD pipeline with automated testing, image build, and deployment, plus structured logging for CloudWatch-compatible observability.

Travel Pi: Network Security Appliance · *Linux, Bash, WireGuard, iptables/UFW, Pi-hole, Tailscale, systemd, hostapd*

- Provisioned and configured a portable Linux network security appliance end to end: WireGuard VPN tunneling, stateful firewall rules via iptables/UFW, DNS-layer filtering with Pi-hole, wireless hotspot bridging via hostapd, and Tailscale mesh networking. Wrote idempotent Bash provisioning scripts and systemd health-check services for self-healing headless operation.

EulogyQuest: Full-Stack Web Application · *React, Node.js/Express, PostgreSQL, Docker, JWT Auth*

- Built a full-stack capstone application with a React frontend and Node.js/Express backend, implementing JWT-based authentication, RESTful API design, and PostgreSQL persistence. Containerized the full stack with Docker Compose for consistent local and production environments.

Logstat · *C++17, CMake, Linux*

- Built a high-performance C++ CLI tool for streaming log analysis, computing per-service latency percentiles and error-rate distributions from large datasets in real time and outputting structured JSON diagnostic reports. Demonstrates systems-level C++ proficiency and performance-oriented design on Linux.

Espresso Compiler · *Java, JFlex, Java CUP*

- Implemented a full compiler pipeline from scratch: hand-written lexer, LALR parser, AST construction with visitor pattern, symbol table management, type checking, and modifier validation with structured error reporting and a comprehensive test harness.

TECHNICAL SKILLS

Languages: Python, C++, Kotlin, Java, JavaScript, Bash, Go, SQL, HTML/CSS

Embedded & Hardware: Embedded Linux, real-time control loops, hardware bring-up, motor controllers and servo drives, sensors, RS-232/RS-485 serial interfaces, system integration

Mobile & UI: Kotlin Multiplatform, Jetpack Compose, MVVM, Retrofit/OkHttp, Koin, kotlinx.serialization, DataStore, React

Cloud & DevOps: AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Terraform, GitHub Actions, Git, CI/CD

Backend & APIs: FastAPI, REST, Node.js/Express, PostgreSQL, SQLAlchemy, Docker Compose

Networking & Security: TCP/IP, DNS, SSH, WireGuard/IPsec, iptables/UFW, VPN, Linux networking

Systems: Linux (Ubuntu/Debian/embedded), systemd, Bash scripting, CMake, GDB, process management

EDUCATION

Georgia Institute of Technology, Atlanta, GA

Spring 2027

M.S. in Computer Science

University of Nevada, Las Vegas, Las Vegas, NV

Graduated May 2025

B.S. in Computer Science